Exam 2

ECE 2201: Circuit Analysis I

Fall 2024

November 2, 2024

1. This exam is closed book, closed notes and a crib sheet has been provided to you with the exam. Print your name below.
2. Show all work on these pages. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit.
3. Show **all units in solutions**, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 90 minutes to work on this exam.

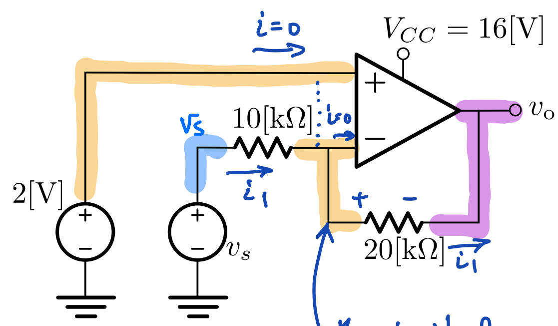
Name: Solution ______________________ ID: ___________

1. _______ /25
2. _______ /25
3. _______ /25
4. _______ /25

Problem 1

For the following circuit, assume an ideal op-amp and

a. Write an expression for v_O in terms of v_S
b. Determine the linear range of v_S



Alternatively: note that
$2[V] - V_o = i_1 \cdot 20[k\Omega]$
so:
$V_o = 2[V] - i_1 \cdot 20[k\Omega]$

through virtual short, node is at 2[V] w/ respect to the reference node.

$$i_1 = \frac{V_S - 2[V]}{10[k\Omega]}$$

$$\begin{aligned}
V_o &= 2[V] - i_1 \cdot 20[k\Omega] \\
&= 2[V] - \frac{(V_S - 2[V])}{10[k\Omega]} \cdot 20[k\Omega] \\
&= 2[V] - 2(V_S - 2[V]) \\
&= 2[V] - 2V_S + 4[V] \\
V_o &= -2V_S + 6[V]
\end{aligned}$$

$$\begin{aligned}
-16[V] &< V_o < +16[V] \\
-16[V] &< -2V_S + 6[V] < 16[V] \\
-22[V] &< -2V_S < 10[V] \\
11[V] &> V_S > -5[V] \\
-5[V] &< V_S < 11[V]
\end{aligned}$$

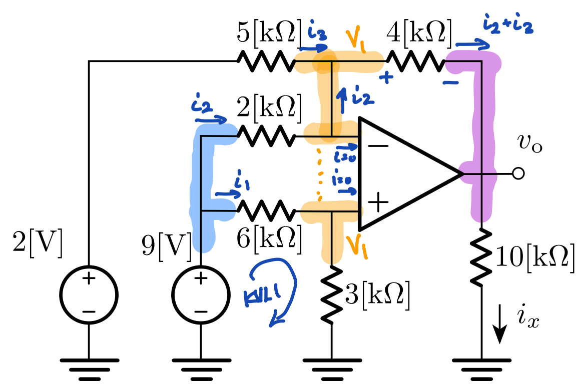
$v_O =$ $-2V_S + 6[V]$

linear range: $-5[V] < V_S < 11[V]$

Problem 2

For the following circuit, assume an ideal op-amp and

a. Find the value of v_O
b. Find the value of i_X



From KVL1:

$$9[V] = i_1 \cdot 6[k\Omega] + i_1\, 3[k\Omega]$$

$$i_1 = 1[mA]$$

$$V_1 = i_1\, 3[k\Omega] = 3[V]$$ → All nodes in orange are at the same potential due to the virtual short.

$$i_2 = \frac{9[V] - V_1}{2[k\Omega]} = \frac{6[V]}{2[k\Omega]} = 3[mA]$$

$$i_3 = \frac{2[V] - V_1}{5[k\Omega]} = \frac{-1[V]}{5[k\Omega]} = -0.2[mA]$$

$$i_2 + i_3 = 2.8[mA]$$

$$v_o = \overbrace{3[V]}^{V_1} - \overbrace{28[mA]\cdot 4[k\Omega]}^{(i_1+i_2)\cdot 4[k\Omega]}$$

$$v_o = -8.2[V]$$

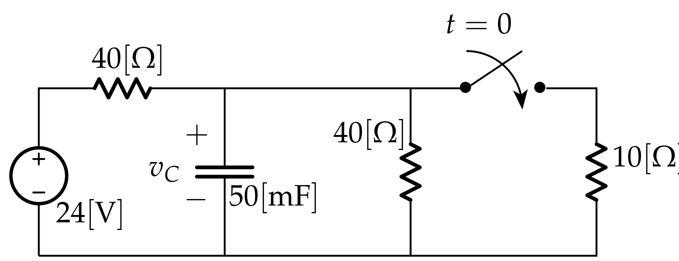
$$i_x = \frac{v_o}{10[k\Omega]} = -0.82[mA]$$

$v_O =$ $-8.2[V]$

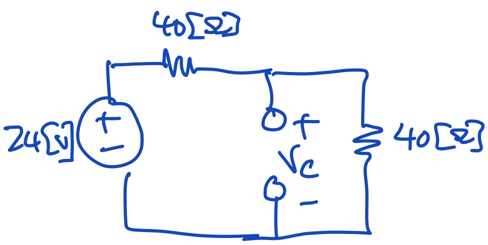
$i_X =$ $-0.82[mA]$

Problem 3

Determine $v_C(t)$ for $t > 0$, where $v_C(t)$ is the voltage across the capacitor. Assume the switch has been open for a long time before $t = 0$. For your answers, write the values of $v_C(0)$, $v_C(\infty)$, and τ, as well as the expression for $v_C(t)$ in the spaces provided below.



$t \leq 0$

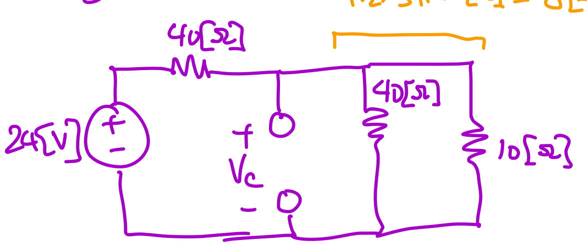


Since a long time has passed, the capacitor is an open and

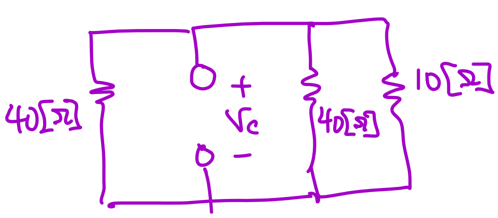
$$V_c = \frac{40[\Omega]}{40[\Omega]+40[\Omega]} \cdot 24[V] = 12[V] = V_c(0)$$

$t \to \infty$

$40[\Omega] \,||\, 10[\Omega] = 8[\Omega]$



$$V_c = 24[V] \cdot \frac{8[\Omega]}{40[\Omega]+8[\Omega]} = 4[V] = V_c(\infty)$$



Req. seen by capacitor

$$40[\Omega] \,||\, 40[\Omega] \,||\, 10[\Omega] = \frac{20 \cdot 10}{30}[\Omega] = \frac{20}{3}[\Omega]$$

$v_C(0) =$ 12[V]

$v_C(\infty) =$ 4[V]

$\tau =$ 333 [ms]

$v_C(t) =$ $4[V] + 8[V]e^{-t/333[ms]}$

$$\tau = \frac{20}{3}[\Omega] \cdot 50 \times 10^{-3}[F]$$

$$= \frac{1000}{3} \cdot 10^{-3}[s]$$

$$= 333\,[ms]$$

Hence:

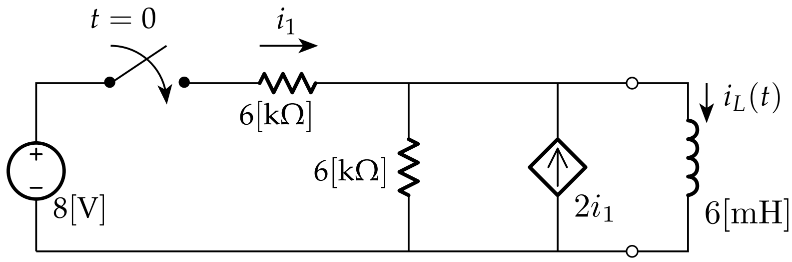
$$V_c = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}$$

$$= 4[V] + 8[V]e^{-t/333[ms]}$$

Problem 4

Determine $i_L(t)$ for $t > 0$, where $i_L(t)$ is the current flowing through the inductor. Assume the switch has been open for a long time before $t = 0$. For your answers, write the values of $i_L(0)$, $i_L(\infty)$, and τ, as well as the expression for $i_L(t)$ in the spaces provided below.

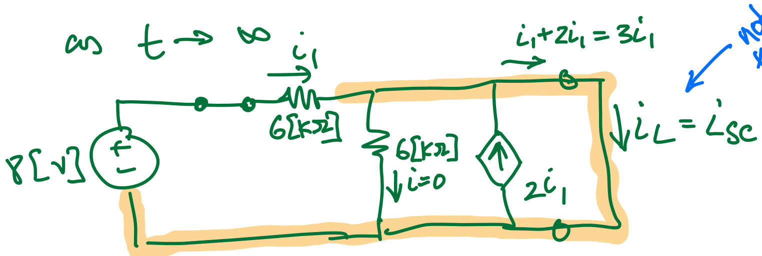
Bonus: Sketch a graph of $i_L(t)$ vs. t, labeling axes and units. How many time constants does it take for the value of $i_L(t)$ to get within 5% of the final value. Show this clearly on the graph.



when $t<0[s]$, inductor is a short. Also, $i_1 = 0$.

hence $i_L(0) = 0$

as $t \to \infty$



note: i_L is also the short-circuit current

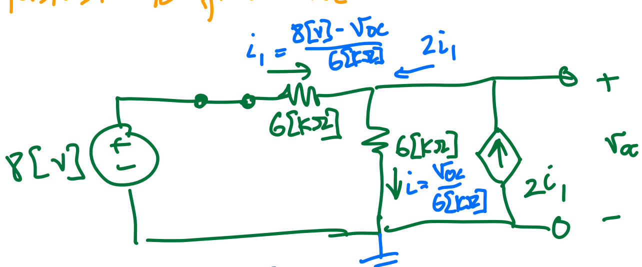
Note that the current through the vertical resistor is 0

so $i_L = i_1 + 2i_1 = 3i_1$

Also, $i_1 = \frac{8[V]}{6[k\Omega]} = \frac{4}{3}[mA]$

so: $i_L(\infty) = i_{sc} = 4[mA]$

To find the time constant, we need the R_{eq}. Since we have i_{sc}, it's probably fastest to find v_{oc}:



$$3i_1 = 3\frac{(8[V]-v_{oc})}{6[k\Omega]} = \frac{v_{oc}}{6[k\Omega]}$$

$$v_{oc} = 3\cdot 8[V] - 3v_{oc}$$

$$4v_{oc} = 24[V]$$

$$v_{oc} = 6[V]$$

So: $R_{eq} = \frac{v_{oc}}{i_{sc}}$

$$= \frac{6[V]}{4[mA]} = \frac{3}{2}[k\Omega]$$

and $\tau = \frac{6[mH]}{\frac{3}{2}[k\Omega]} = 4[\mu s]$

$i_L(0) =$ 0

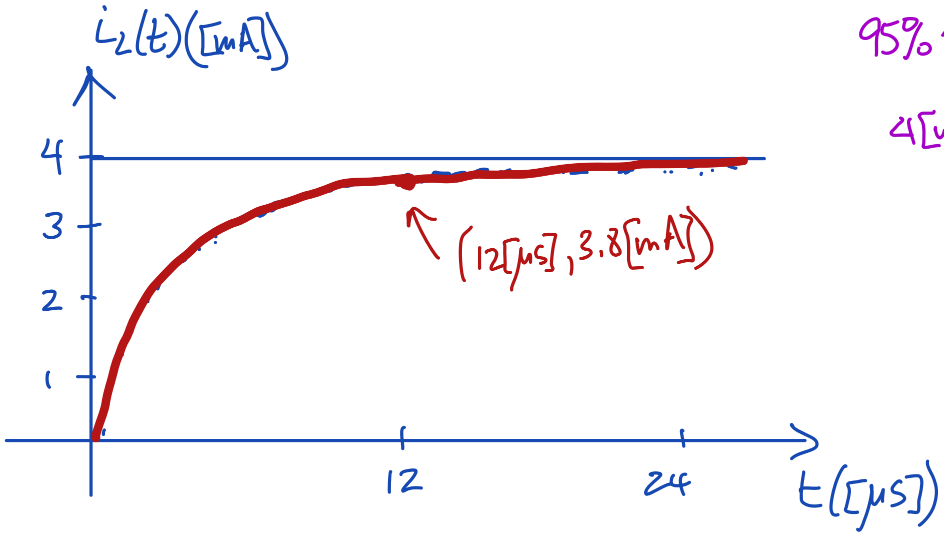
$i_L(\infty) =$ 4[mA]

$\tau =$ 4[μs]

$i_L(t) = 4[mA]\left(1 - e^{-t/4[\mu s]}\right)$

$$i_L(t) = i_L(\infty) + (i_L(0) - i_L(\infty))e^{t/\tau}$$

Bonus:



$$95\% \cdot 4[\text{mA}] = 3.8[\text{mA}]$$

$$4[\text{mA}]\left(1 - e^{-t/[4\mu s]}\right) = 3.8[\text{mA}]$$

$$-e^{-t/4[\mu s]} = \frac{3.8}{4} - 1$$

$$e^{-t/4[\mu s]} = \frac{0.2}{4} = 0.05$$

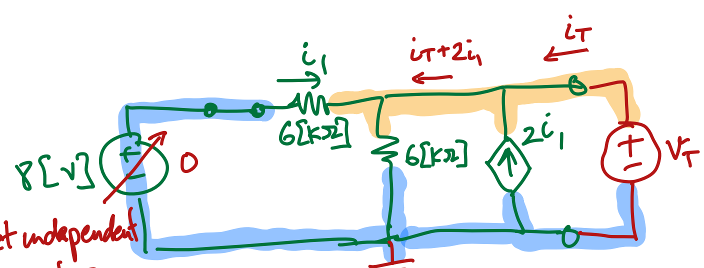
$$-\frac{t}{4[\mu s]} = \ln(0.05) = -3$$

$$t = 3 \cdot \overbrace{4[\mu s]}^{\tau} = 12[\mu s]$$

$$= 3 \cdot \tau$$

3 time constants

Alternative method to finding $R_{eq.}$: Test source



set independent source to zero: voltage source becomes a short

$$V_T = (i_T + 2i_1 + i_1)\, 6[\text{k}\Omega]$$

$$= (i_T + 3i_1)\, 6[\text{k}\Omega]$$

also $\quad i_1 = \dfrac{0 - V_T}{6[\text{k}\Omega]} = -\dfrac{V_T}{6[\text{k}\Omega]}$

so: $\quad V_T = \left(i_T - \dfrac{3V_T}{6[\text{k}\Omega]}\right) 6[\text{k}\Omega]$

$$V_T = 6[\text{k}\Omega]\, i_T - 3V_T$$

$$4V_T = 6[\text{k}\Omega]\, i_T$$

$$\frac{V_T}{i_T} = R_{eq.} = \frac{6[\text{k}\Omega]}{4} = \frac{3}{2}[\text{k}\Omega]$$